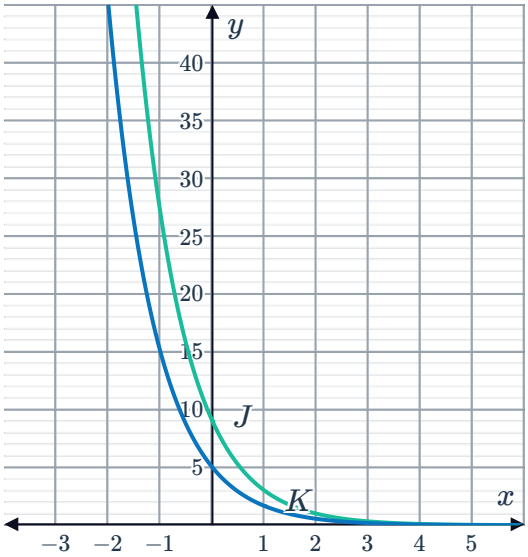
# Comparison of exponential functions

**1**

**a** Decreasing

**b**




**c** As $x \to \infty$, functions $J, K \to 0$.

**d** Dilate $y = 3^{-x}$ vertically by a factor of $5$.

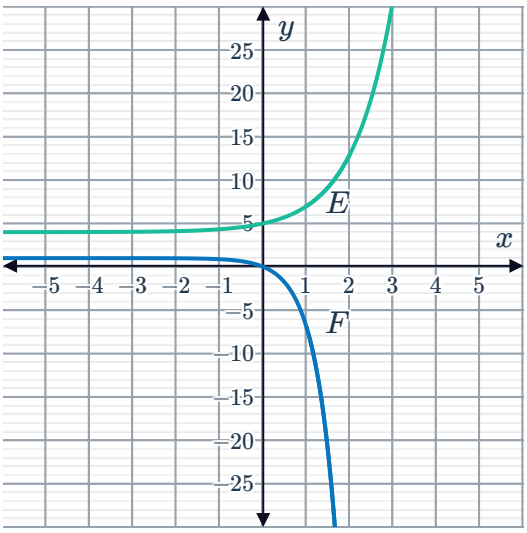
**2**

**a** Once

**b** Function $P$

**3**

**a**




**b** Zero times

**c** As $x \to \infty$, the function $E \to \infty$ and function $F \to -\infty$.

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## Applications

**4**

**a** 8 million

**b**

**i** 7130 million **ii** 9286 million

**5**

**a**

**i** $A = P(1.053)^t$ **ii** $A = P(1.0125)^{4t}$

**b** $5.09\%$

**c** Bank A

**6**

**a** Sale price of property

**b** $5.3\%$

**c** $-5.3\%$

**7**

**a** $M = 60\,000(1.032)^t$

**b** $P = 50\,000(1.061)^t$

**c** Mohamad's salary will be $\$1153.51$ greater than Elizabeth's.

**d** Elizabeth's salary

**8**

**a** $30$

**b** $40\%$

**c** $100\%$

**d** The population of bacteria in the dark environment is increasing at a faster rate than the population of bacteria in the sunlit environment.

**9**

**a** $P = 90 + 5t$

**b** $Q = 90 \times (1.04)^t$

**c** The current company

**d** Opening his own business

**10**

**a** $P = 100(0.85)^t$

**b** Fabian

**c** No, neither Fabian or Zainab will completely forget the content. There is a horizontal asymptote at $y = 0$ and so the recall percentage for both will never reach zero.

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**11** **a** Yes, there will be a food shortage.
We can derive the equation $F = 3\,374\,000\,(1.02)^n$ to represent the growth of the population, and the equation $P = 2\,410\,000\,(1.022)^n$ to represent the number of people the food supply will feed.
We can see that the food production is increasing, but at a slower rate than the population's rate of increase, so the population will eventually exceed the amount that can be fed.

**b** We can draw the graph of both functions, using technology to find any points of intersection. If the $y$-value of the population is greater than the $y$-value of the population that can be fed for the same value of $n$ then there will be a food shortage.

**12**

**a** $179$

**b** $132$

**c** Blair

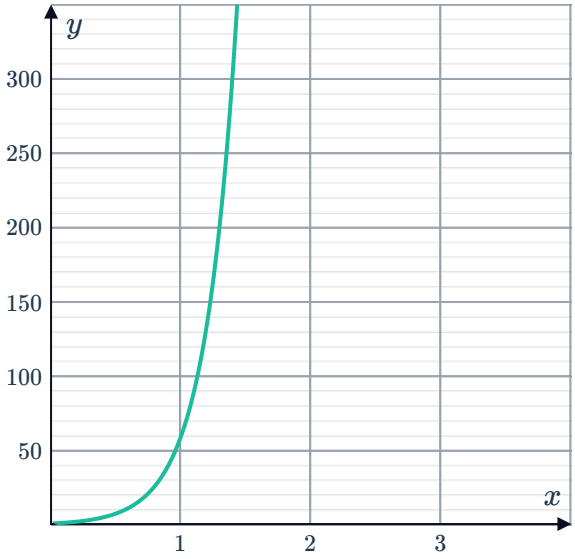
**d** $70\%$

**e** If you have a high sugar diet, your blood sugar level peaks at greater amounts and decreases more rapidly than a low sugar diet.

**13**

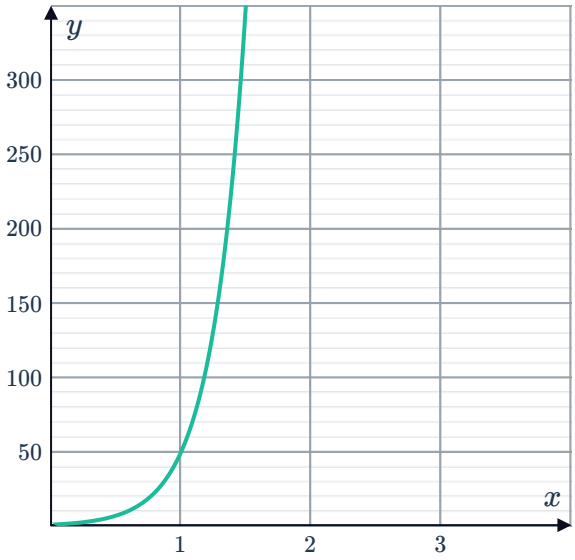
**a** The population of bacteria $J$ increases faster than the population of $K$.

**b**




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C




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